Past Meets Present: A Tale of Two Visions

TTIC 41000

Instructor: Anand Bhattad

Lecture 1: Introduction



Introduction: Anand

- I am a Research Assistant Professor at TTIC since Sept 2023
- Before that, a PhD student at UIUC with David A. Forsyth
- Research Interest: Computer Vision, Computer Graphics and Computational Photography
- Current Research Focus
  - Generative Models
    - Exploring Knowledge, Image Editing
  - Image-based Rendering
    - Object Compositing, Image-based Relighting
  - Neural Radiance Fields
    - Interpretable and Controllable Fields, Understanding building blocks of NeRF

Today's Agenda

- Why this class?
- Requirements and logistics
- How is the class going to be?
- Marr's Vision

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It was the best of times,
it was the worst of times,
it was the age of wisdom,
it was the age of foolishness,
it was the epoch of belief,
it was the epoch of incredulity,
it was the season of light,
it was the season of darkness,
it was the spring of hope,
it was the winter of despair.

- Charles Dickens,
A Tale of Two Cities

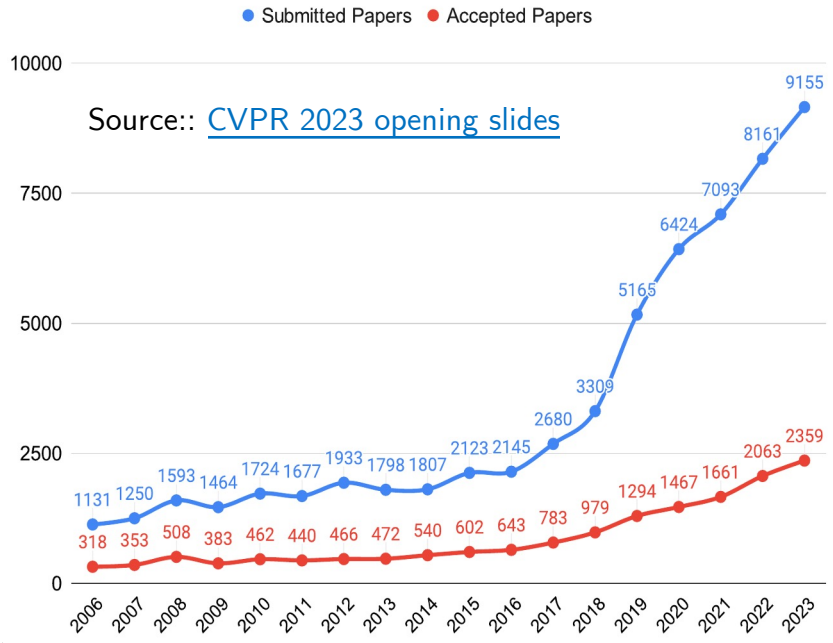


“It was the best of times…”



Source: Sora (Open AI)

“It was the best of times…”





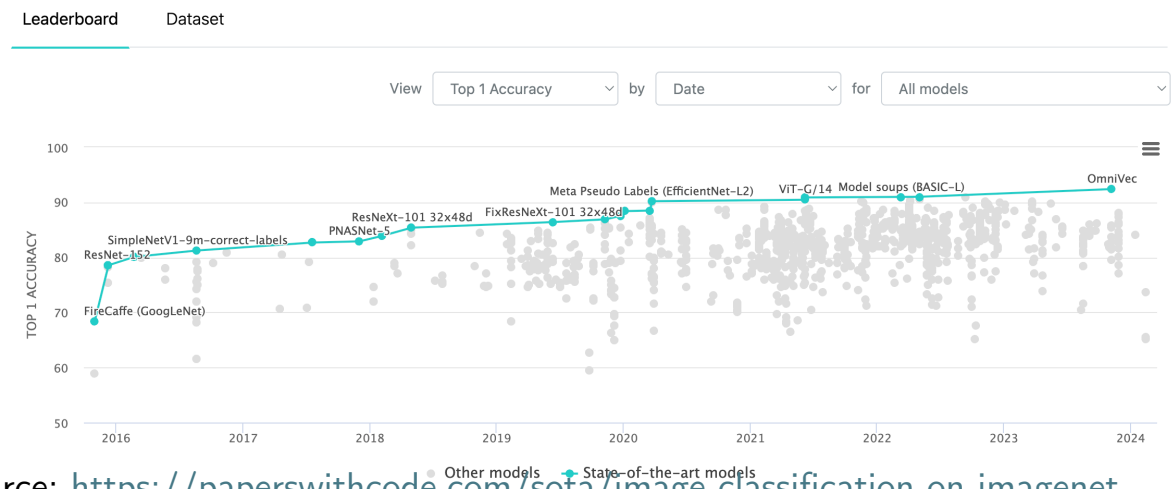
Source: Bram Grooten on ~~Twitter~~(X) from NeurIPS 2023

Categories ▾ English ▾

	Publication	h5-index	h5-median
1.	Nature	467	707
2.	The New England Journal of Medicine	439	876
3.	Science	424	665
4.	IEEE/CVF Conference on Computer Vision and Pattern Recognition	422	681
5.	The Lancet	368	688
6.	Nature Communications	349	456
7.	Advanced Materials	326	415
8.	Cell	316	503
9.	Neural Information Processing Systems	309	503
10.	International Conference on Learning Representations	303	563

Source: Google Scholar

Image Classification on ImageNet



Source: https://paperswithcode.com/sota/image-classification-on-imagenet

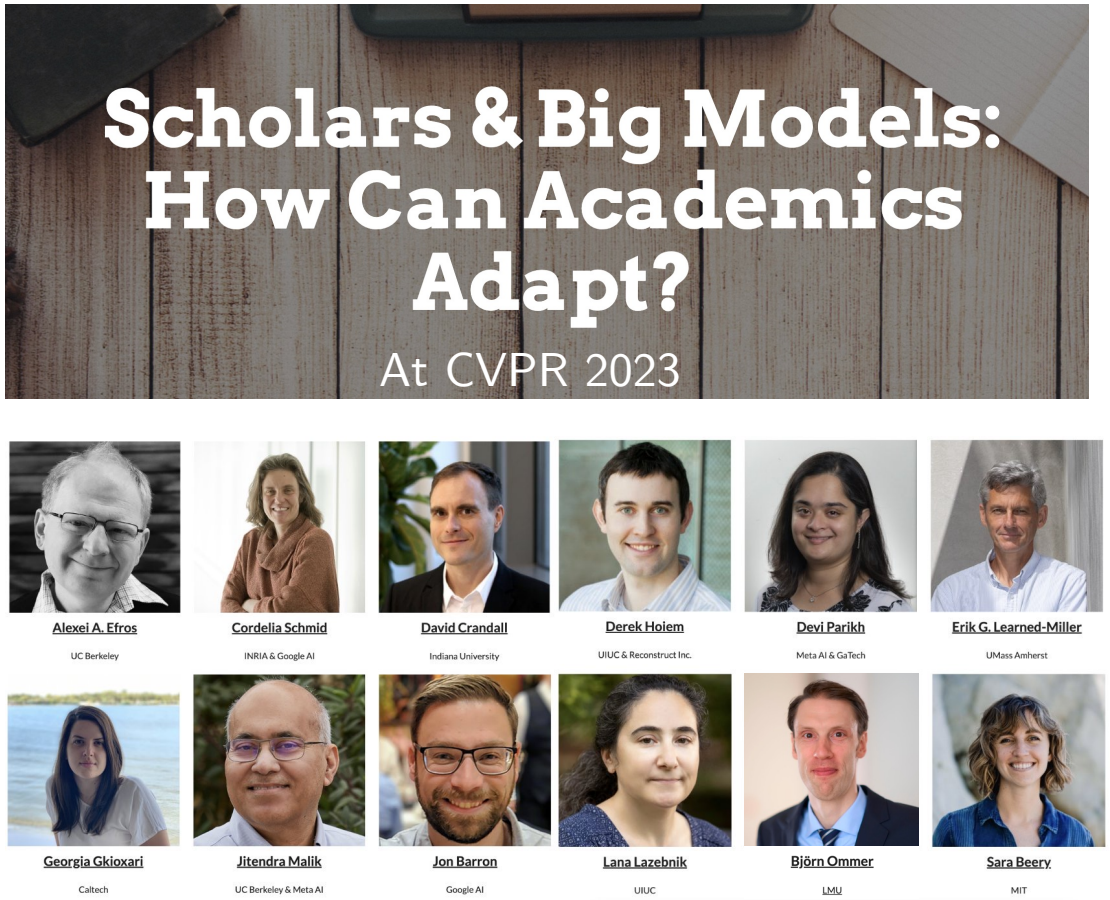
“It was the worst of times...”

- “I have no compute”
- “No one cares about my small model”
- “My research is no longer relevant. For real.”
- “I can’t access big models!”
- “What did OpenAI train DALL-E on?”
- “Is it ok to use this data?”
- “Will I have no job?”

Source: Scholars & Big Models Opening slides
(from a survey before the workshop):



“We have everything before us, we have nothing before us...”





https://sites.google.com/view/academic-cv/

https://gkioxari.github.io/Tutorials/iccv2023/

“Change begets change. Nothing propagates so rapidly”



Source: https://sites.google.com/view/retrocv

“Change begets change. Nothing propagates so rapidly”

Computer Vision: Looking Back to Look Forward

Svetlana Lazebnik

IRIM Visiting Faculty Fellow Mini-Course

January 28 - February 6, 2020



https://slazebni.cs.illinois.edu/spring20/

CS 598 LAZ: Computer Vision: What Will Stand the Test of Time?



Meeting time: W F 11AM - 12:15PM, 1320 DCL

Instructor: **Svetlana Lazebnik** (slazebni -at- illinois.edu)

TA: Aiyu Cui (aiyucui2)

Contents of webpage: reading list, requirements, schedule

Announcements, discussion: Campuswire

Submission of deliverables, grades: Canvas

https://slazebni.cs.illinois.edu/fall23/

“It was the age of wisdom, it was the age of foolishness”

Past Meets Present: A Tale of Two Vision

- Engage with the foundational concepts and seminal works that have shaped computer vision
- Explore recent advancements and SOTA approaches in computer vision, appreciating the progress made possible by building upon the solid foundations.
- Bridge the gap between historical wisdom and modern approach, identifying dots that connect the past and present, for a better future

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This Class: Past Meets Present

- https://anandbhattad.github.io/teaching/past_meets_present/

- Topics: https://docs.google.com/spreadsheets/d/11NWzULs6WsVaIL0t_qH4AN_47bIF1x0rE3c3zglrfSk

- Slack: https://join.slack.com/t/slack-yji5371/shared_invite/zt-2evvhpolh-gQUQYPyWQ0Kbj6ypDgysTQ



Requirements

- Presentations (2x): 60%
  - An historical foundation paper and a modern approach paper
- Participation: 20%
- Project: 20%
  - Use Old Ideas with Modern Tools

Deciphering 'What' and 'Where' Visual Pathways from Spectral Clustering of Layer-Distributed Neural Representations

Xiao Zhang* University of Chicago zhang7@uchicago.edu
David Yunis* TTI-Chicago dyunis@ttic.edu
Michael Maire University of Chicago mmaire@uchicago.edu

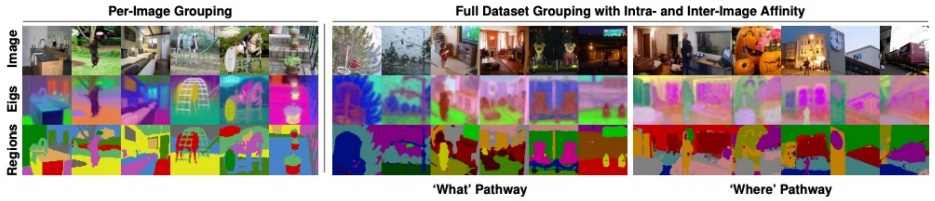


Figure 1. Our novel optimization procedure, resembling spectral clustering, leverages features throughout layers of a pre-trained model to extract dense structural representations of images. Shown are results of applying our method to Stable Diffusion [51]. ***Left:*** Analyzing internal feature affinity for a single input image yields region grouping. ***Right:*** Extending the affinity graph across images yields coherent dataset-level segmentation and reveals 'what' (object identity) and 'where' (spatial location) pathways, depending on the feature source.

Zhang*, Yunis*, Maire. CVPR 2024

Region-Based Representations Revisited

Michal Shlapentokh-Rothman[1*] Ansel Blume[1*] Yao Xiao[1] Yuqun Wu[1]
Sethuraman T V[1] Heyi Tao[1] Jae Yong Lee[1] Wilfredo Torres[2] Yu-Xiong Wang[1]
Derek Hoiem[1,2]
[1] University of Illinois at Urbana-Champaign
[2] Reconstruct

Abstract

We investigate whether region-based representations are effective for recognition. Regions were once a mainstay in recognition approaches, but pixel and patch-based features are now used almost exclusively. We show that recent class-agnostic segmenters like SAM can be effectively combined with strong self-supervised representations, like those from DINOv2, and used for a wide variety of tasks, including semantic segmentation, object-based image retrieval, and multi-image analysis. Once the masks and features are extracted, these representations, even with linear

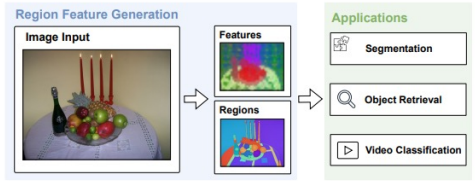


Figure 1. Our framework revisits the use of region features for downstream applications. We generate region features by first segmenting an image, extracting image features, then pooling the image features across the region masks.

Shlapentokh-Rothman*, Blume* et al. CVPR 2024

Class Format: Two papers per class

- Historical Foundational Paper (25 mins + 5 mins for questions)
  - Introduce topic or problem statement with precise formal definition
  - Discuss specifics of the paper
  - Bonus: one or two slides summarize key developments till modern approach
- Modern Approach paper (25 mins + 5 mins for questions)
  - Discuss specifics of the paper
  - Make an effort to connect modern approach with foundational paper
  - Bonus: Future direction combining old ideas with modern approach
- Discussion led by me (15 mins)
  - Questions collected via survey before every class
  - Open audience questions

Presentation Requirements

- **Signup – by the start of next class, Mar 21st!**
- Practice presentation (30% of grade)
- Slides (30% of grade)
- In-class presentation (30% of grade)
- Peer grading (10% of grade)

Project Requirements

- Proposal (10% of project grade) – due on April 8th
- Progress report or first draft (10% of grade) – due on April 29th
- Project Presentation (30% of grade) – on May 16th (last day of class)
- Final report (50% of grade) – May 16th, end of the day

Who Should Attend This Class?

- Expect a small/close setting with engaging discussions
- Introductory Computer Vision Knowledge is recommended
  - For newcomers, recommend watching Greg's videos here.
- Enrollment is must for those who wish to take the class for credit
- Auditing is possible with permission
  - active participation is still expected
- You can also register for pass/fail or equivalent
  - Then no need to do course project (but I would highly recommend it!)
  - must present papers and contribute to discussions

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Example: Face Recognition

- Identify or verifies a person's identity using their facial features
- Decades of research and used to be a huge part of computer vision
- Applications: security systems, surveillance, and personal device authentication

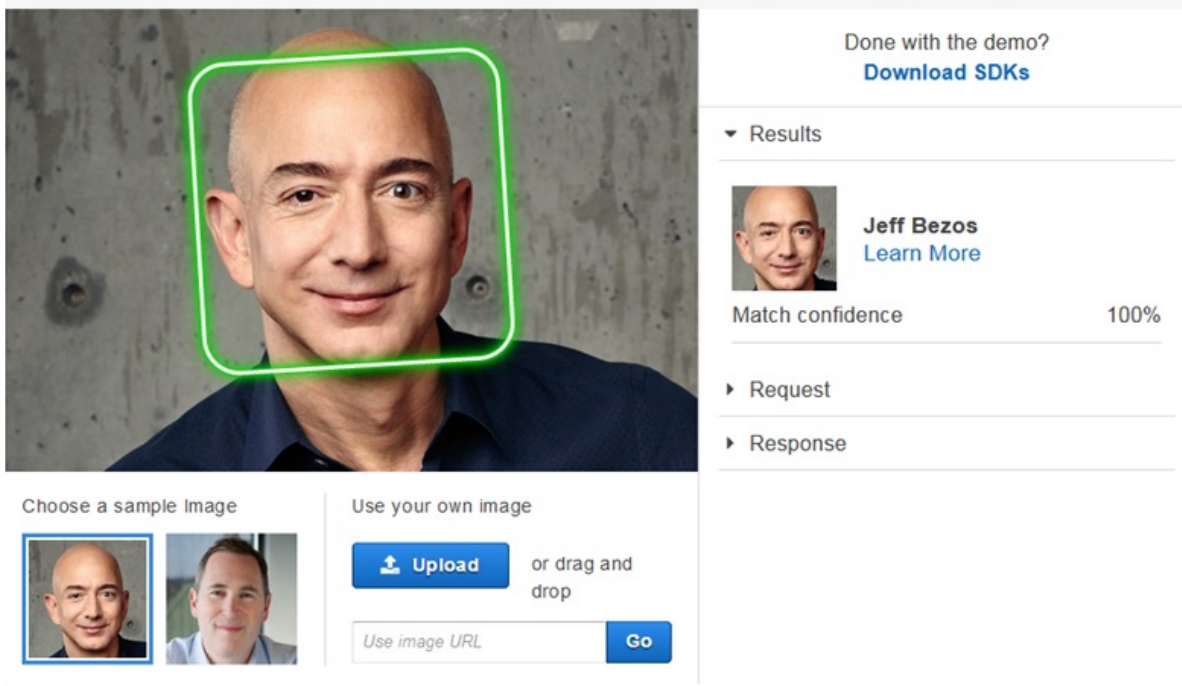


Image Source

Eigenfaces

- Straightforward and computationally efficient approach
- Assumes that most face images lie on a low-dimensional subspace
- Dimensionality reduction using PCA
  - Represents faces using a small set of basis images called “eigenfaces”
  - Eigenfaces (u_1 u_2 u_3... u_k) are the eigenvectors of the covariance matrix
  - First k eigenvectors (k<<d, where d is original image dimension) are sufficient to represent a face

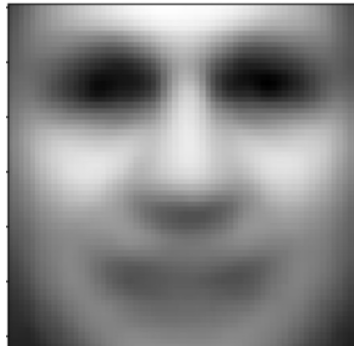
Turk & Pentland (1991)

Eigenfaces

- Eigenfaces capture the most significant variations among the face images in the dataset
- Each eigenface represents a different facial feature or characteristics



Training Dataset



Mean Face



Eigenfaces

Turk & Pentland (1991)

Images from here

Eigenface as Representation

- Each face image in the dataset is represented as a linear combinations of eigenfaces
- Coefficients are computed using the dot product between the face image and each eigenface



50 Eigenfaces

100 Eigenfaces

250 Eigenfaces

1000 Eigenfaces

[Turk & Pentland (1991)]

Images from [here]

Recognition and Matching

- A query face image is projected onto the eigenface subspace to obtain its coefficients
- The coefficients are compared with those of the faces in the database using a similarity metric (e.g., Euclidean distance)
- The face in the database with the closest match (minimum distance) is considered the recognized individual

Turk & Pentland (1991)

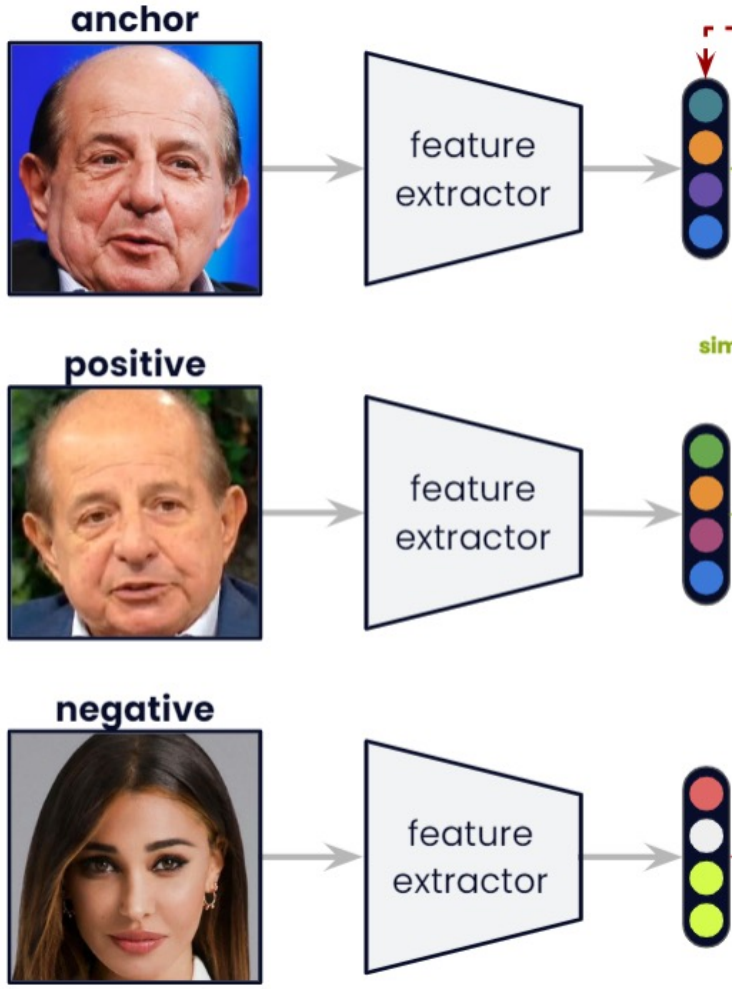
FaceNet



Florian Schroff, Dmitry Kalenichenko and James Philbina (2015)

Image from here

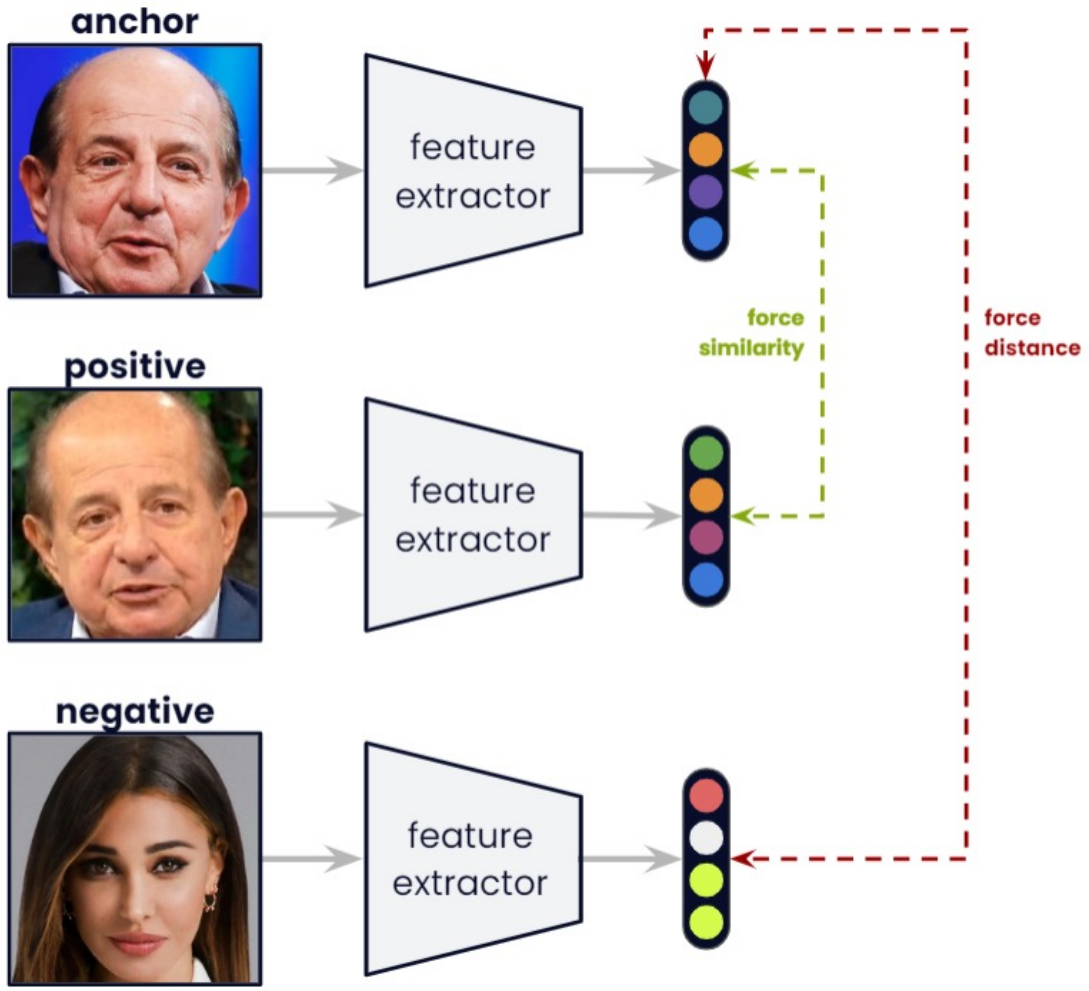
FaceNet



Florian Schroff, Dmitry Kalenichenko and James Philbina (2015)

Image from here

FaceNet



Triplet Loss

Florian Schroff, Dmitry Kalenichenko and James Philbina (2015)

Image from here

Comparing Eigenfaces and FaceNet

Similarities

- Compact Representation

Differences

- Eigenfaces is a self-supervised and FaceNet is supervised
- Eigenfaces is generative model and FaceNet is discriminative model
- Eigenfaces is based on global structure and FaceNet learns hierarchical features from low-level to high-level representations

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Slides from Lana Lazebnik

Machine Perception of 3Dimensional Solids

- 1963: Roberts Ph.D. thesis at MIT

MACHINE PERCEPTION OF THREE-DIMENSIONAL SOLIDS

by

LAWRENCE GILMAN ROBERTS

Submitted to the Department of Electrical Engineering on May 10, 1963, in partial fulfillment of the requirements for the degree of Doctor of Philosophy.



1937– 2018
Wikipedia bio

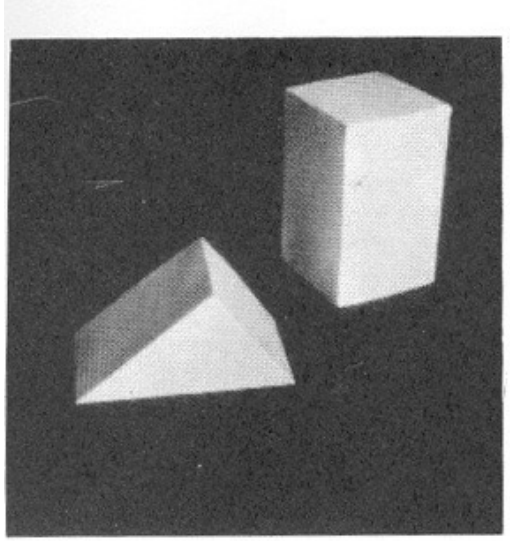
Slide from Lana Lazebnik

CHAPTER I

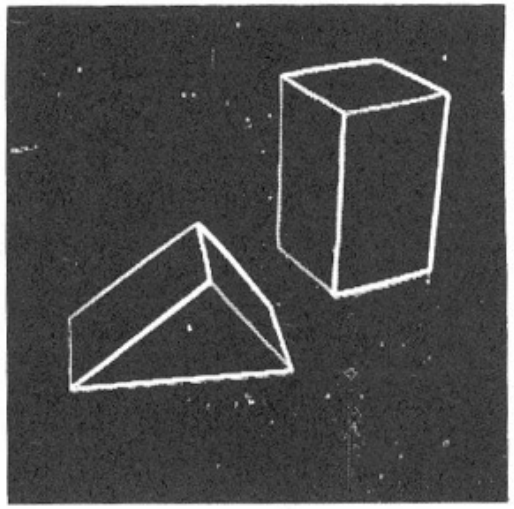
INTRODUCTION

The problem of machine recognition of pictorial data has long been a challenging goal, but has seldom been attempted with anything more complex than alphabetic characters. Many people have felt that research on character recognition would be a first step, leading the way to a more general pattern recognition system. However, the multitudinous attempts at character recognition, including my own, have not led very far. The reason, I feel, is that the study of abstract, two-dimensional forms leads us away from, not toward, the techniques necessary for the recognition of three-dimensional objects. The perception of solid objects is a process which can be based on the properties of three-dimensional transformations and the laws of nature. By carefully utilizing these properties, a procedure has been developed which can not only identify objects, but also determines their orientation and position in space.

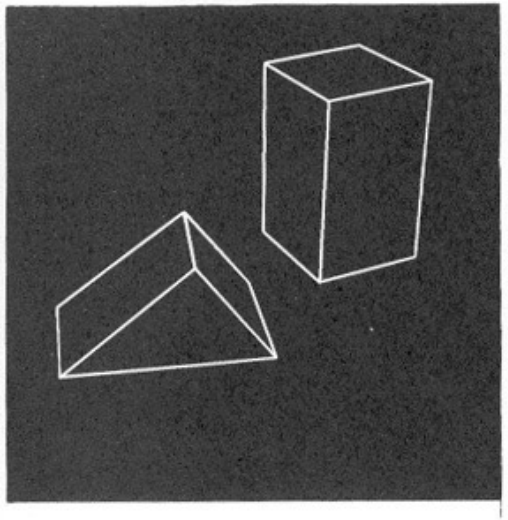
Slide from Lana Lazebnik



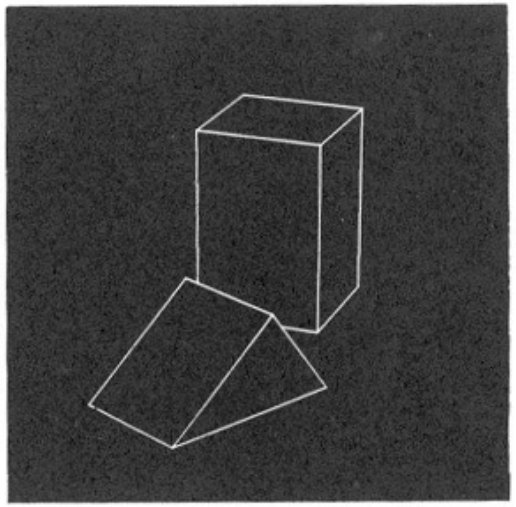
(a) Original picture.



(b) Differentiated picture.



(c) Line drawing.



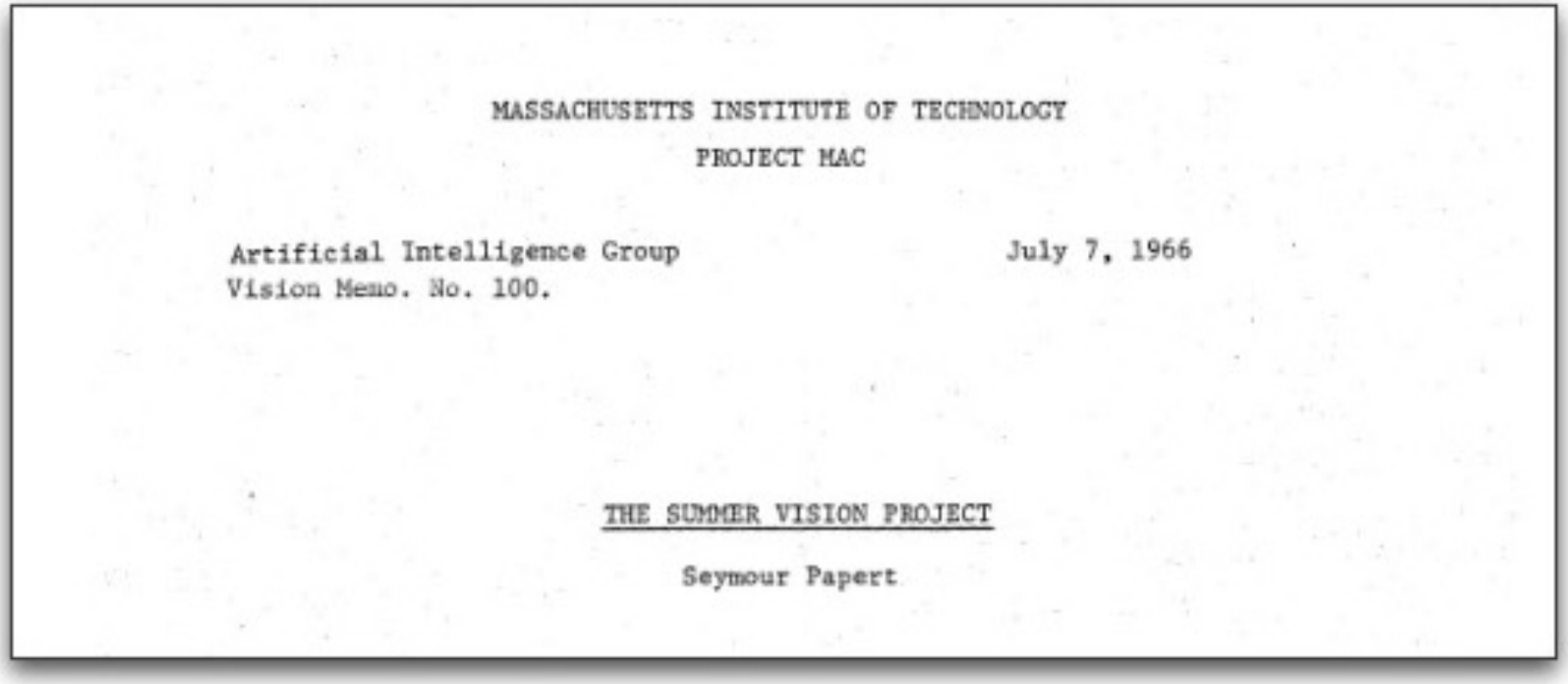
(d) Rotated view.

From the abstract:

"It is assumed that a photograph is a projection of... **known three-dimensional models**... These assumptions enable a computer to obtain a reasonable, three-dimensional description from the edge information in a photograph by means of a topological, mathematical process."

Slide from Lana Lazebnik

MIT Summer Vision Project



MASSACHUSETTS INSTITUTE OF TECHNOLOGY

PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT

Seymour Papert

MASSACHUSETTS INSTITUTE OF TECHNOLOGY

PROJECT MAC

Artificial Intelligence Group — July 7, 1966
Vision Memo. No. 100.

THE SUMMER VISION PROJECT

Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

Slide from Lana Lazebnik

Subgoal for July

Analysis of scenes consisting of non-overlapping objects from the following set:

balls

bricks with faces of the same or different colors or textures

cylinders.

Each face will be of uniform and distinct color and/or texture. Background will be homogeneous.

Extensions for August

The first priority will be to handle objects of the same sort but with complex surfaces and backgrounds, e.g. cigarette pack with writing and bands of different color, or a cylindrical battery.

Then extend class of objects to objects like tools, cups, etc.

See Lana Lazebnik's Computer Vision: What will stand test of time? Lecture 2 for History

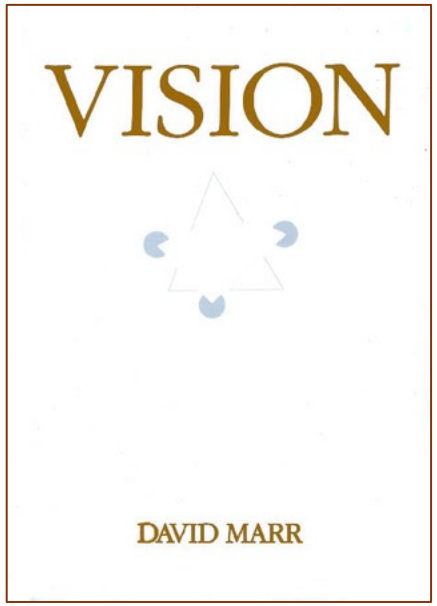
Slide from Lana Lazebnik

David Marr (1945-1980)

- Ph.D. in theoretical neuroscience, Cambridge, 1969
  - Models of the cerebellum (1969), neocortex (1970), hippocampus (1971)
- Joined MIT AI Lab in 1973, became professor of psychology in 1977
  - Stereo algorithms (with Tommaso Poggio), 1976-79
  - 3D object representation (with Keith Nishihara), 1978
  - Edge detection (with Ellen Hildreth), 1980
- Posthumous book: Vision (1982)



Bio



Full text

In December 1977, certain events occurred that forced me to write this book a few years earlier than I had planned. Although the book has important gaps, which I hope will soon be filled, a new framework for studying vision is already clear and supported by enough solid results to be worth setting down as a coherent whole.

Slide from Lana Lazebnik

Marr's motivation (ch. 1)

- Vision is *hard*

> The first great revelation was that the problems are difficult. Of course, these days this fact is a commonplace. But in the 1960s almost no one realized that machine vision was difficult. The field had to go through the same experience as the machine translation field did in its fiascoes of the 1950s before it was at last realized that here were some problems that had to be taken seriously. The reason for this misperception is that we humans are ourselves so good at vision. The notion of a feature detector was well established by Barlow and by Hubel and Wiesel, and the idea that extracting edges and lines from images might be at all difficult simply did not occur to those who had not tried to do it. It turned out to be an elusive problem: Edges that are of critical importance from a three-dimensional point of view often cannot be found at all by looking at the intensity changes in an image. Any kind of textured image gives a multitude of noisy edge segments; variations in reflectance and illumination cause no end of trouble; and even if an edge has a clear existence at one point, it is as likely as not to fade out quite soon, appearing only in patches along its length in the image. The common and almost despairing feeling of the early investigators like B.K.P. Horn and T.O. Binford was that practically anything could happen in an image and furthermore that practically everything did.

Slide from Lana Lazebnik

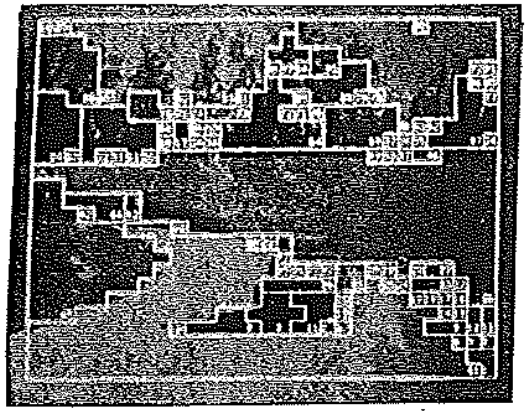
Marr's motivation (ch. 1)

- Vision is *hard*
- We may not be able to figure out the right solution right away, but at least we should start by establishing a sound methodology
  - Marr explicitly considered and rejected low-level neurophysiology, empirical "hacking", and blocks world simplification



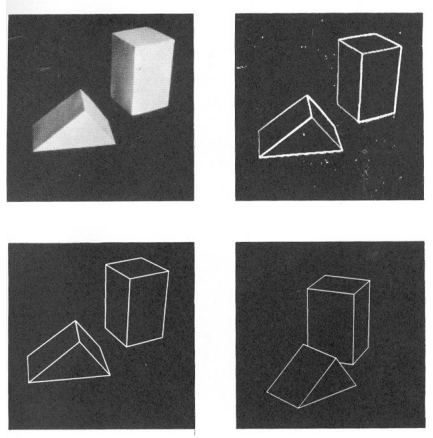
Hubel & Wiesel (1959)

(source)



(B-2) Output of the non-semantic weakest boundary melted first region grower.

Yakimovsky & Feldman (1973)



Roberts (1963)

Slide from Lana Lazebnik

Towards an *information processing* theory of vision

Computational theory

What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?

Slide from Lana Lazebnik

Towards an *information processing* theory of vision

Computational theory	Representation and algorithm
What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?	How can this computational theory be implemented? In particular, what is the representation for the input and output, and what is the algorithm for the transformation?

Slide from Lana Lazebnik

Towards an *information processing* theory of vision

Computational theory	Representation and algorithm	Hardware implementation
What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?	How can this computational theory be implemented? In particular, what is the representation for the input and output, and what is the algorithm for the transformation?	How can the representation and algorithm be realized physically?

Figure 1–4. The three levels at which any machine carrying out an information-processing task must be understood.

Slide from Lana Lazebnik

Computational theory description of vision

- What should be the goal of vision?
  - "Vision is a process that produces from images of the external world a description that is useful to the viewer and not cluttered with irrelevant information."
- What should be the input?
  - "In the case of human vision, the initial representation is in no doubt – it consists of **arrays of image intensity values** as detected by the photoreceptors in the retina."
- What should be the output?
  - "[The purpose of vision is] **building a description of the shapes and positions of things** from images... It also tells about the illumination and about the reflectances of the surfaces that make the shapes – their brightnesses and colors and visual textures – and about their motion. But these things seemed secondary; they could be hung off a theory in which **the main job of vision was to derive a representation of shape**."

Slide from Lana Lazebnik

Proposed algorithmic pipeline

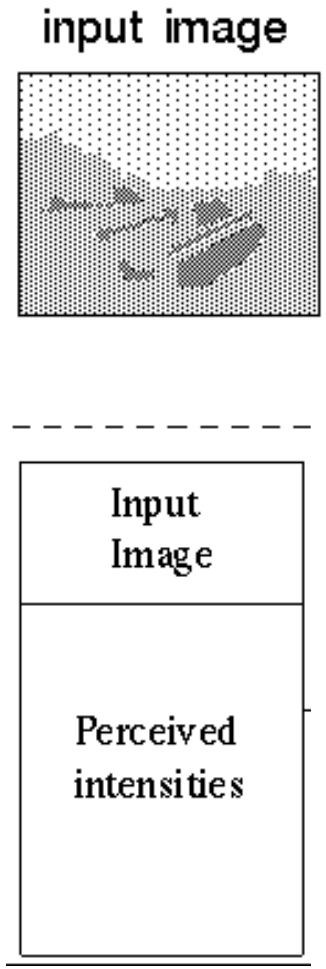


Image source

Slide from Lana Lazebnik

Proposed algorithmic pipeline

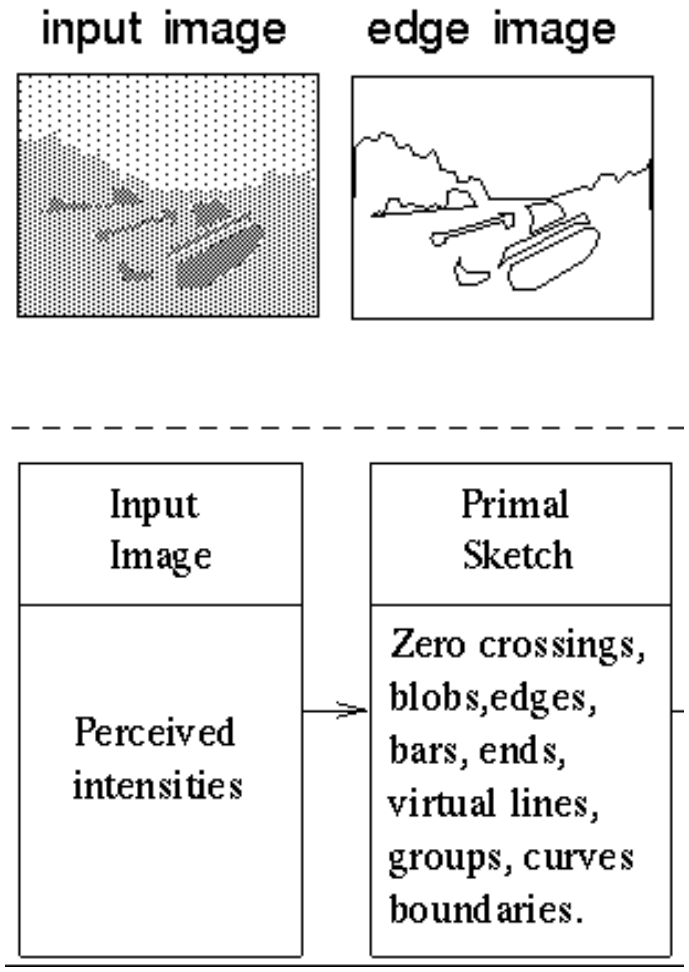


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Slide from Lana Lazebnik

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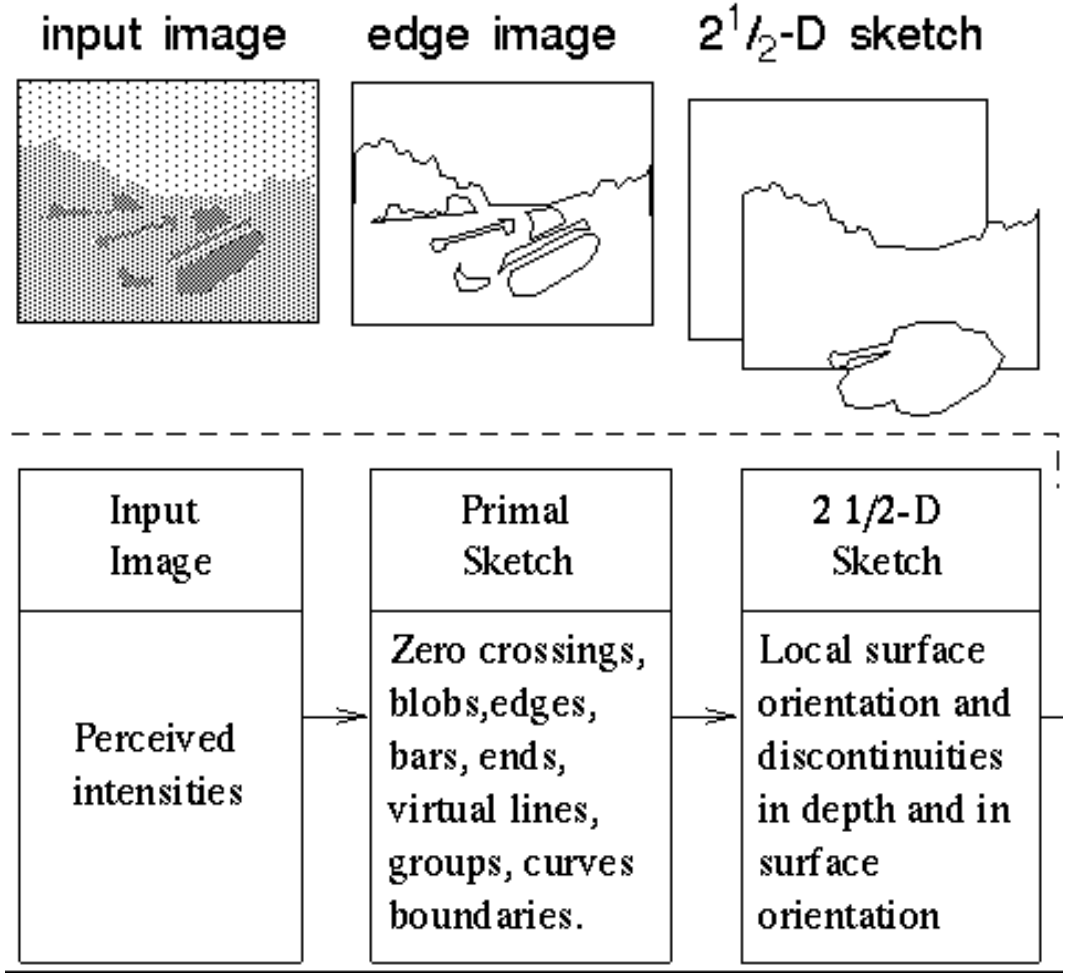


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Slide from Lana Lazebnik

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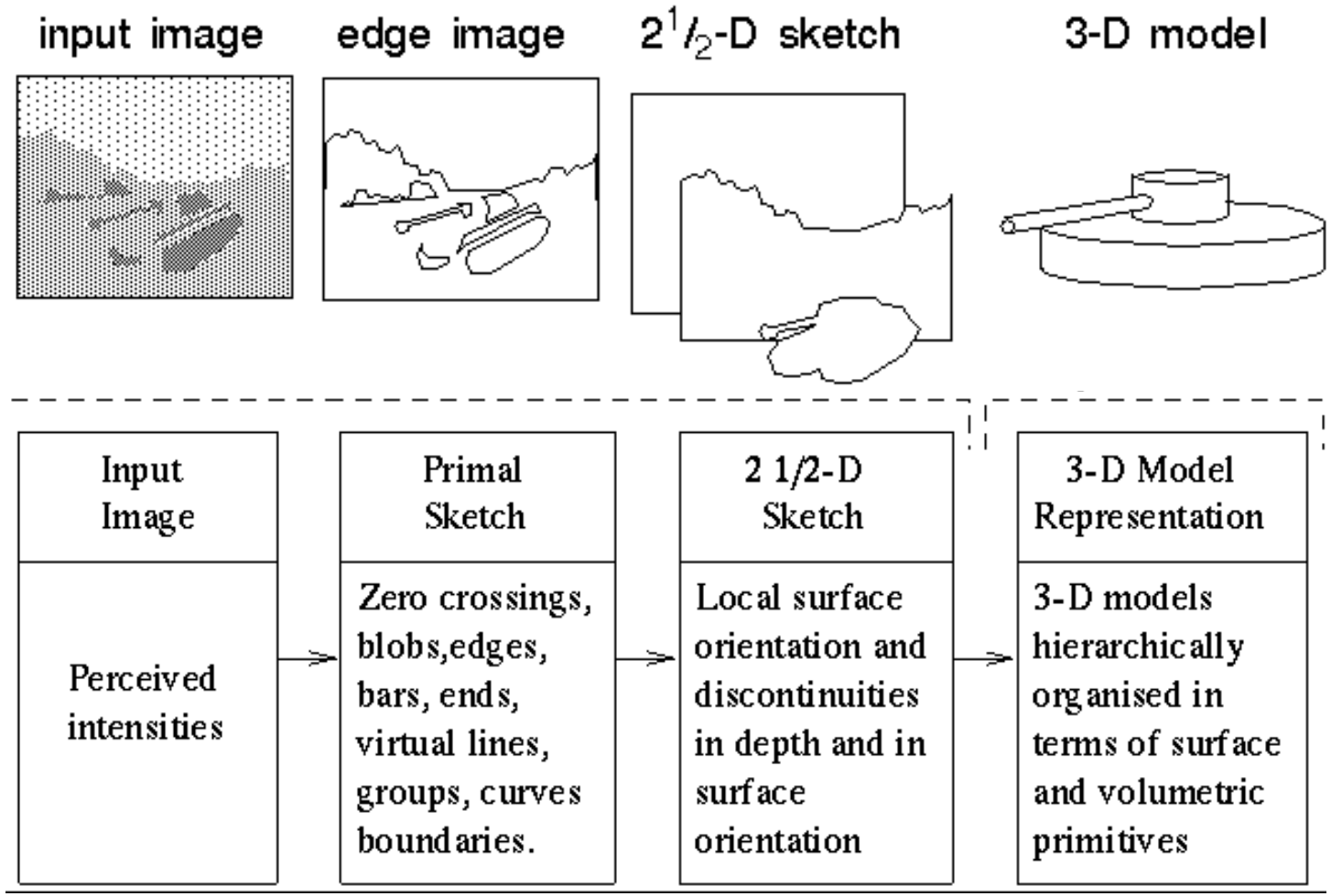


Image source

Slide from Lana Lazebnik

So, what's the big deal?

- Marr's book was a major milestone
  - Critical summary of key developments in study of human and computer vision to date
  - Unprecedented attempt at a unified account of the entire visual system
- Computational framework was very appealing to computer vision researchers from a "software engineering" perspective
  - Abstraction, modularity, feedforward pipeline
- Theories meshed well with the dominant computer vision paradigms
  - Vision as "inverse graphics" or "inverse optics"
  - Emphasis on recovery of general-purpose 3D representations composed of simple geometric primitives
  - Convenient division of vision problems into "low-level", "mid-level", and "high-level"

Special issue dedicated to Marr: Perception 41(9), 2012

Slide from Lana Lazebnik

What about the bad stuff?

- Marr's approach have not panned out either on the human or the computer vision side (did not consider feedback)
- Principles of modularity and feedforward processing don't hold for human vision
  - P. Churchland, V.S. Ramachandran, and T. Sejnowski, A critique of pure vision, 1994
- Humans do not recover veridical, task-independent 3D representations
  - W. Warren, Does This Computational Theory Solve the Right Problem? Marr, Gibson, and the Goal of Vision, Perception 41(9), 2012
- Marr dismissed statistical approaches, did not even consider learning
- Even the goals, inputs, and outputs of a vision system are very much open to question (see embodied agent literature)

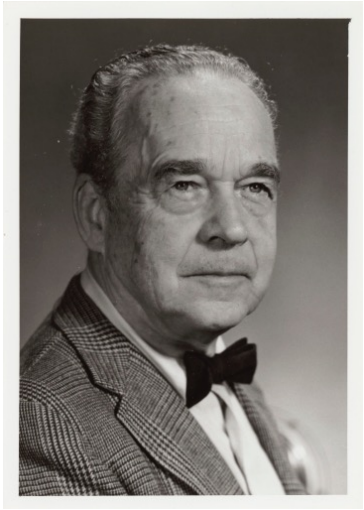
Slide from Lana Lazebnik

Was it only Marr who came up with “Theory of Perception”?

Marr, Gibson, Cavanagh, Koenderink of Many



David Marr (1945-1980)



James Jerome Gibson (1904-1979)



Patrick Cavanagh (b. 1947)



Jan Johan Koenderink (b. 1943)



Giorgione, The Four Philosophers

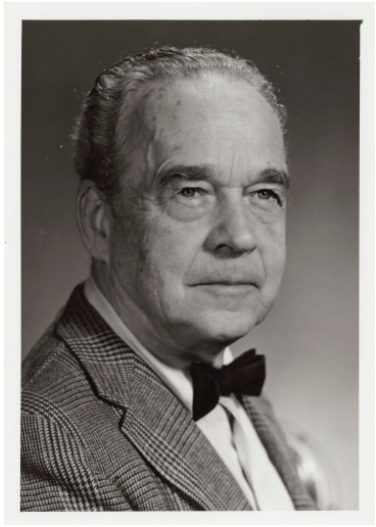
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Summary of 4 Views on Perception

- Marr, Gibson, Cavanagh and Koenderink all asked about the nature of vision and came up with different answers



"Vision is a computational process that transforms the retinal image into an objective representation of 3D shape. Vision is inverse optics."



"There is no computation. There is no retinal image. There are no representations. There is no 3D shape. There is only direct pickup of ecologically relevant variants and invariants. Vision is in the world, not the observer."



"Perception is not a passive process of recording the world but an active process of constructing representations of the world that are shaped by our knowledge, expectations, and goals."



"There is no objective world, only the observer's *umwelt* (surrounding world). Thus, vision cannot be in the world but is a creative act of the observer."

Slide adapted from Lana Lazebnik

Inspirational quote

“The past is never dead. It's not even past.”

– William Faulkner

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